FINN CALLAHAN

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PROFILE

- Computational Biology Ph.D. student at University of California, Berkeley working in Dr. Rasmus Nielsen's lab
- Currently working on computational methods development for ecological modeling using data from ancient environmental DNA
- Interested in pedagogy, mentorship, and equity-oriented teaching practices

EDUCATION

PhD University of California, Berkeley (in progress) Expected December 2026
Computational Biology

BA Scripps College, GPA 3.9 May 2021
Mathematical/Computational Biology
Major through Harvey Mudd College

PUBLICATIONS

Note: some publications appear under Fiona Margaret Callahan

De Sanctis, B.†, Rouillard, A.†, …**Callahan, F.**, … & Willerslev, E. 3.7 million year old environmental DNA from High Arctic Canada. (in review).

Callahan, F., Kjær, K. H., Ramsøe, A. D., Willerslev, E., & Nielsen, R. Reproducibility and normalization in ancient environmental DNA. (in preparation).

Kjær, K. H.†, Ruter, A. H.†, Menendez-Serra, M.†, Vogel, N. A.†, Ramsøe, A. D.†, Farnsworth, W. R.†, Siggaard-Andersen, M.-L.†, Huang, Z.†, **Callahan, F.†**, Mesny, F.†, … Willerslev, E. (in review). Environmental DNA reveals Reykjavík's human and ecological history.

Callahan, F. M.†, & Evensen, C.† (2026). Co-occurrence networks can preserve emergent properties of ecological communities. *bioRxiv*, 2026-04. DOI:10.64898/2026.04.15.718781. (also, in review).

Li, J. K., Lim, W., **Callahan, F.**, Raskin, L. Y., Lemmon-Kishi, M., & Nielsen, R. (2026). STEM-LM: Spatio-Temporal Ecological Modeling via Masked Language Model for Joint Species Distribution. *bioRxiv*, 2026-05.

Callahan, F. M., Li, J. K., & Nielsen, R. (2025). Challenges in detecting ecological interactions using sedimentary ancient DNA data. *Environmental DNA*, *7*(2), e70067. DOI:10.1002/edn3.70067.

†*Joint first authors*

RESEARCH EXPERIENCE

Network modeling for community ecology Spring 2025 - Present
UC Berkeley
Advisor: Dr. Rasmus Nielsen

- Designed simulations to benchmark species interaction network modeling tools.
- Explored how co-occurrence networks and species interaction networks are related through generalized Lotka-Volterra models.

Ecological community modeling using ancient eDNA Spring 2022 - Present
UC Berkeley
Advisor: Dr. Rasmus Nielsen

- Developed mechanistic simulation framework for ecological data from ancient environmental DNA (aeDNA)
- Analysis of aeDNA data from Iceland (in progress)
- Development of new data analysis tools for aeDNA (in progress)
- Development of rigorous statistical frameworks for normalization and use of replicates for aeDNA data

Time series modeling of phytoplankton blooms in the San Francisco Bay Spring 2022
UC Berkeley (first year PhD rotation project)
Advisor: Dr. Perry de Valpine

- Rotation project (10 weeks) analyzing time series data to predict phytoplankton blooms in the San Francisco Bay

Disease ecology in bees Fall 2021
UC Berkeley (first year PhD rotation project)
Advisor: Dr. Mike Boots

- Rotation project (10 weeks) building a simulation of disease transmission in bees

Connecting molecular differences in TTN gene to muscle performance 2020 - 2021
WM Keck Science Department, Claremont Colleges
Advisor: Dr. Findley Finseth

- Undergraduate senior thesis research
- Characterized genetic differences in the protein Titin to connect them to within-species differences in muscle performance
- Adapted annotation tool for a region of the giant sarcomeric protein titin

Differential gene expression in high-running versus control mice Summer 2019
Differential gene expression in pair-bonded versus naïve zebra finch Spring 2020
WM Keck Science Department, Claremont Colleges
Advisors: Dr. Findley Finseth, Dr. Jenna Monroy

- Performed RNA extraction for RNA-seq and qPCR from mouse muscle tissue
- Analyzed RNA-seq data from mouse muscle tissue and zebra finch brain tissue
- Performed a comparative analysis of computational tools for RNA-seq differential expression analysis

TEACHING AND MENTORSHIP

Interim Lecturer, Department of Integrative Biology, UC Berkeley

- Instructor of record for *Human Biological Variation* (IB35AC; ~400 students; Spring 2027)
- Managed five graduate student instructors
- Modified syllabus and lesson plans to center active learning and student engagement in a large lecture setting

Graduate Student Instructor, Department of Integrative Biology, UC Berkeley

- Led discussion sections for *Human Biological Variation* (IB35AC; ~80 students; Spring 2026)
- Taught *Berkeley Connect in Computational Biology* (40 students; Fall 2023 & Spring 2024)
- Designed original syllabi and lesson plans; coordinated accessibility accommodations; graded writing assignments and creative projects

Math Instructor, Mount Tamalpais College, San Quentin Rehabilitation Center

- Volunteer math instructor for college-level and college-preparatory courses for incarcerated students at San Quentin Rehabilitation Center (Summer 2024 - Present)
- Designed and taught lessons; created assignments and exams for adult students

Research Mentorship

- Mentored 6 undergraduate research assistants in computational biology projects (2022 - Present)
- Designed independent research projects and conducted weekly mentorship meeting

Mentorship & Outreach

- Computational Biology Mentorship Committee (2023 - Present): coordinating professional development workshops and one-on-one peer mentorship for early-career PhD students
- Workshop leader, Expanding Your Horizons conference (2023 - 2026): one-day conference introducing middle school girls and gender-diverse youth to STEM careers through hands-on workshops
- Volunteer mentor, Be A Scientist program: guided 7th grade students in 6-week modules designing and conducting independent experiments

Pedagogical Training

- Completed workshops in inclusive classroom design, accessibility, and evidence-based teaching practices
- Completed pedagogy and mentorship courses at UC Berkeley

COURSEWORK AND SKILLS

- **Computer Science**: Data Structures and Program Development, Algorithms, Introduction to Computer Science, Principles of Computer Science
  - **Programming languages**: Python (advanced), R (advanced), Bash (advanced), Java, C++
  - **Lab server administrator**: managed technical support for two Ubuntu servers for the Nielsen lab
- **Statistics**: Introduction to Statistical Computing (graduate level), Hierarchical Statistical Modeling in Environmental Science (graduate level), Statistical models: Theory and Applications (graduate level), Stochastic Processes, Introduction to Statistics at an Advance Level (graduate level), Introduction to Probability at an Advance Level (graduate level)
- **Ecology and Biology**: Network approaches to infer the resilience of ecological communities, Genomics and Bioinformatics, Biostatistics, Molecular Genetics, Ecology and Environmental Biology, Introduction to Mathematical/Computational Biology, Advanced Computational Biology, Evolutionary Genomics, Topics in Biochemistry and Molecular Biology, Molecular/Cellular Biology Lab, Experimental Biology Lab
  - **Lab skills:** RNA extractions, PCR, qPCR, primer design, gel electrophoresis, bacterial transformations
- **Mathematics**: Number Theory, Real Analysis, Linear Algebra, Discrete Mathematics, Multivariable Calculus

SERVICE

CCB Diversity, Equity, and Inclusion committee, UC Berkeley	(Fall 2023 – Fall 2026)
CCB peer mentorship committee, UC Berkeley	(Fall 2023 – Spring 2026)
CCB seminar organizing committee, UC Berkeley	(Fall 2023 – Spring 2024)
CCB retreat planning committee, UC Berkeley	(Fall 2022 – Fall 2023)

HONORS AND AWARDS

Student Leadership and Service Award, Center for Computational Biology	Spring 2026
Best Senior Thesis Award, Harvey Mudd College Biology Dept.	Spring 2021
Johnson Summer Research Fellowship, Scripps College	Summer 2020